

## Capital Project Case Study

This case study considers the expected costs and benefits to Jiranna Healthcare resulting from a decision to design a centralized nurse triage line. This triage line would assist Jiranna patients in providing self-care and/or with seeking care at an urgent care facility in lieu of more expensive after-hours care in the emergency room.

### Summary

Jiranna Healthcare's main facility is home to more than 80 onsite specialty and surgery clinics, employing over 5,000 staff. In addition to the main hospital, Jiranna Healthcare has 11 satellite clinics, which can contain healthcare services, such as pediatrics, family medicine, and mental health etc. These facilities (the hospital plus outlying clinics) serve a total enrollee population of 97,000.

Currently, Jiranna Healthcare's centralized call center schedules patient appointments for each of the 11 satellite clinics, handling an average of 1,500 to 2,000 calls daily with a staff of 20. Patients routinely have difficulty obtaining urgent or acute care in a timely fashion. Additionally, in three out of four cases, the majority of Jiranna Healthcare's healthcare centers are unable to meet access standards. These access issues have a secondary effect on the call center, which experiences a much higher call rate because members have to call back multiple times to find available appointments. The existing process leads to overutilization of emergency departments for urgent care and non-emergent concerns. In addition, patient satisfaction has steadily declined as a result of the continued lack of access to healthcare services.

To address this problem, there is a proposal to implement a centralized nurse triage line, an offsite phone center that would be staffed by registered nurses with a multitude of specialties, including emergency room (ER) nurses, critical care, surgical, and even some nurse practitioners. These nurses are able to offer callers medical advice encompassing the treatment of fevers, wound care, and emergent conditions, such as chest pain. The nurses are trained to triage conditions to the appropriate level of care, be that at home, at an urgent care center, or at an emergency department.

The major cost impact is the increased salary requirement for the phone center staff, which will entail approximately 33 multi-discipline employees, based on workload and enrollment data. Additional elements of the proposal include hiring an information technology (IT) specialist to manage the triage line's computer system and facility renovations. The main benefit of this proposal is the projected cost reductions in patient care as a result of moving non-emergent care out of the expensive emergency-room setting.

**Assignment**

The “Capital Project Case Study, Part 2” spreadsheet provides cash flow data (costs and benefits) for the proposal. Download and save this Excel spreadsheet in your working Excel Workbook, and use the information provided to complete the following:

1. Calculate the cash inflows and outflows for each year.
2. Evaluate the capital project by calculating the following metrics:
  - a. Net present value (NPV)
  - b. Internal rate of return (IRR)
  - c. Modified internal rate of return (MIRR)
  - d. Payback period
  - e. Discounted payback period
3. In a 1- to 2-page report, provide your recommendation with a rationale, as to whether the project is acceptable, assuming Jiranna Healthcare has a corporate policy of not accepting projects that take more than 3.5 years to pay for themselves, and assuming an 11% cost of capital.